

Lab 1 – Implementation of Lexical Analyzer

Aim

To design and implement a simple lexical analyzer in C that identifies keywords, identifiers, operators, and special symbols from a given source program.

Algorithm

1. Start the program.
2. Read the source program as a line of text.
3. Define a list of C keywords.
4. Scan the input character by character.
5. If a character is alphabetic or underscore, collect characters until a non-alphanumeric character occurs.
6. Compare the collected token with the keyword list.
 - If matched, print Keyword.
 - Otherwise print Identifier.
7. If the character is an operator (+ - * / = < > %), print Operator.
8. If the character is a delimiter or special symbol (; , () { } []), print Special Symbol.
9. Ignore spaces, tabs, and newlines.
10. Repeat until the end of input.
11. Stop.

Code

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char *keywords[] = {
    "int", "float", "char", "double", "if", "else",
    "while", "for", "return", "void", "break", "continue"
};

int isKeyword(char str[]) {
    int n = sizeof(keywords) / sizeof(keywords[0]);
    for (int i = 0; i < n; i++) {
        if (strcmp(str, keywords[i]) == 0)
            return 1;
    }
    return 0;
}

int isOperator(char ch) {
```

```

        return (ch == '+' || ch == '-' || ch == '*' || ch == '/' ||
                ch == '=' || ch == '<' || ch == '>' || ch == '%');
    }

```

```

int isSpecial(char ch) {
    return (ch == ';' || ch == ',' || ch == '(' || ch == ')' ||
            ch == '{' || ch == '}' || ch == '[' || ch == ']');
}

```

```

int main() {
    char input[500];
    char token[100];
    int i = 0, j;

    printf("Enter source code:\n");
    fgets(input, sizeof(input), stdin);

    while (input[i] != '\0') {
        if (isspace(input[i])) {
            i++;
            continue;
        }

        if (isalpha(input[i]) || input[i] == '_') {
            j = 0;
            while (isalnum(input[i]) || input[i] == '_') {
                token[j++] = input[i++];
            }
            token[j] = '\0';

            if (isKeyword(token))
                printf("%s -> Keyword\n", token);
            else
                printf("%s -> Identifier\n", token);
        }
        else if (isOperator(input[i])) {
            printf("%c -> Operator\n", input[i]);
            i++;
        }
        else if (isSpecial(input[i])) {
            printf("%c -> Special Symbol\n", input[i]);
            i++;
        }
        else {
            i++;
        }
    }
}

```

```
    return 0;  
}
```

Sample Input

```
int a = b + 10;
```

Sample Output

```
int -> Keyword  
a -> Identifier  
= -> Operator  
b -> Identifier  
+ -> Operator  
; -> Special Symbol
```

Result

Thus the lexical analyzer was implemented successfully and it identified keywords, identifiers, operators, and special symbols from the given input.